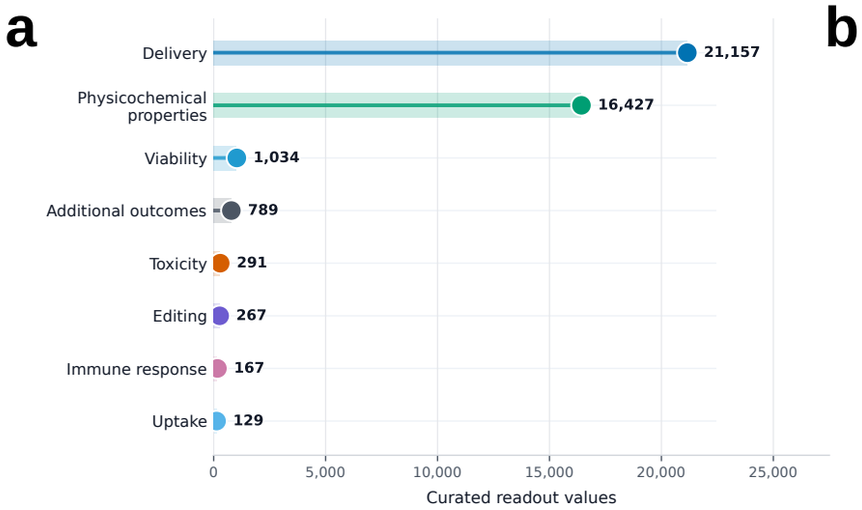
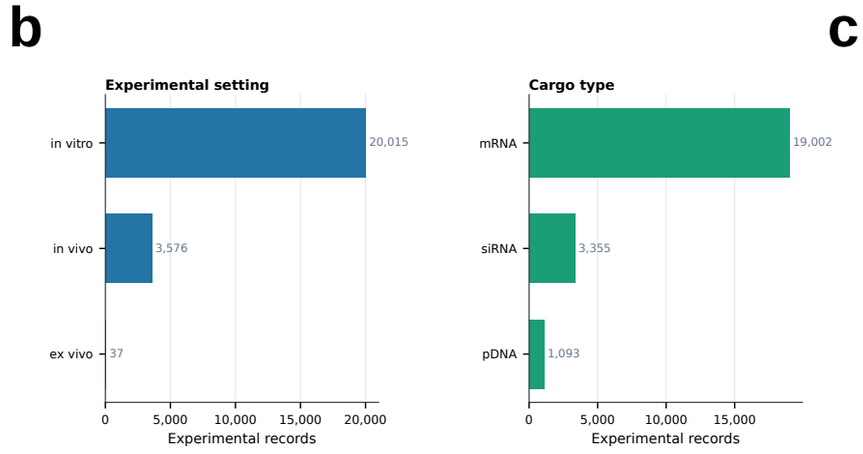
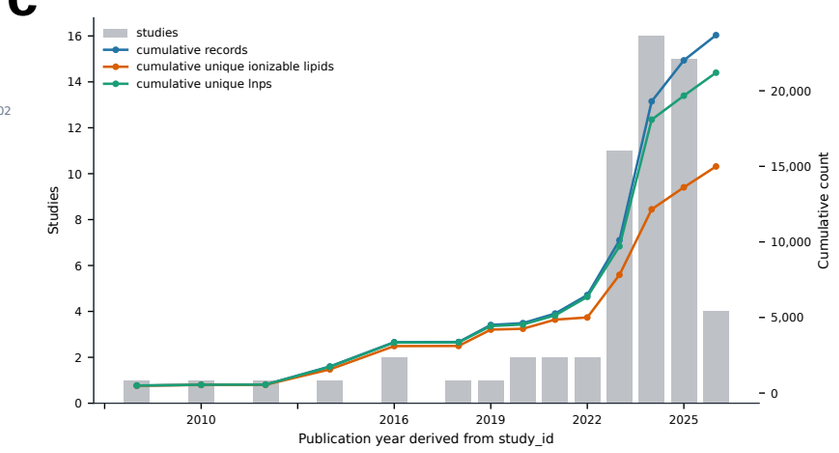
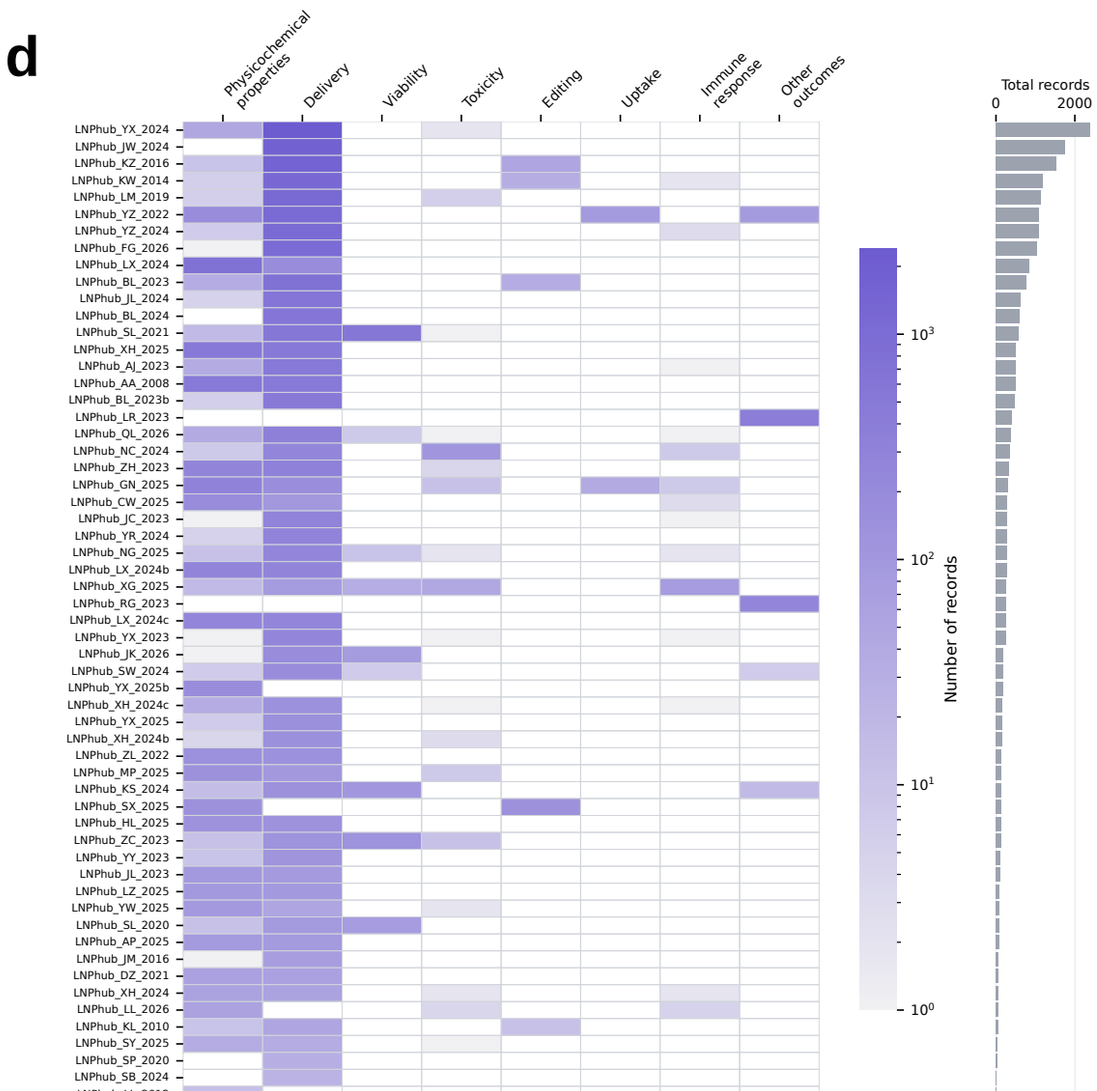
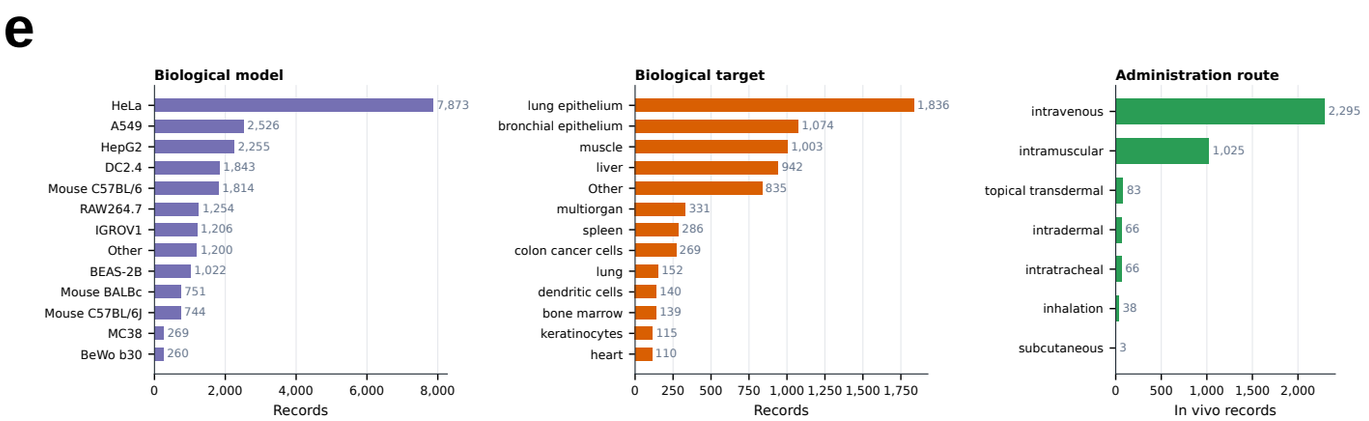
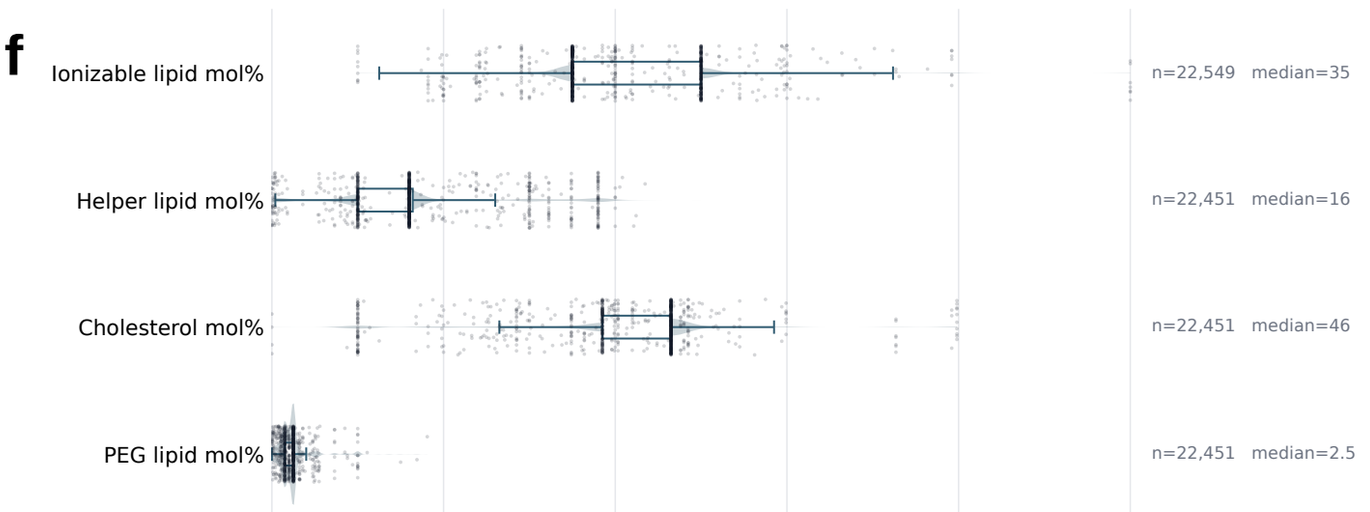

a
Delivery
21,157
Physicochemical properties
16,427
Viability
1,034
Additional outcomes
789
Toxicity
291
Editing
267
Immune response
167
Uptake
129
0
5,000
10,000
15,000
20,000
25,000
Curated readout values

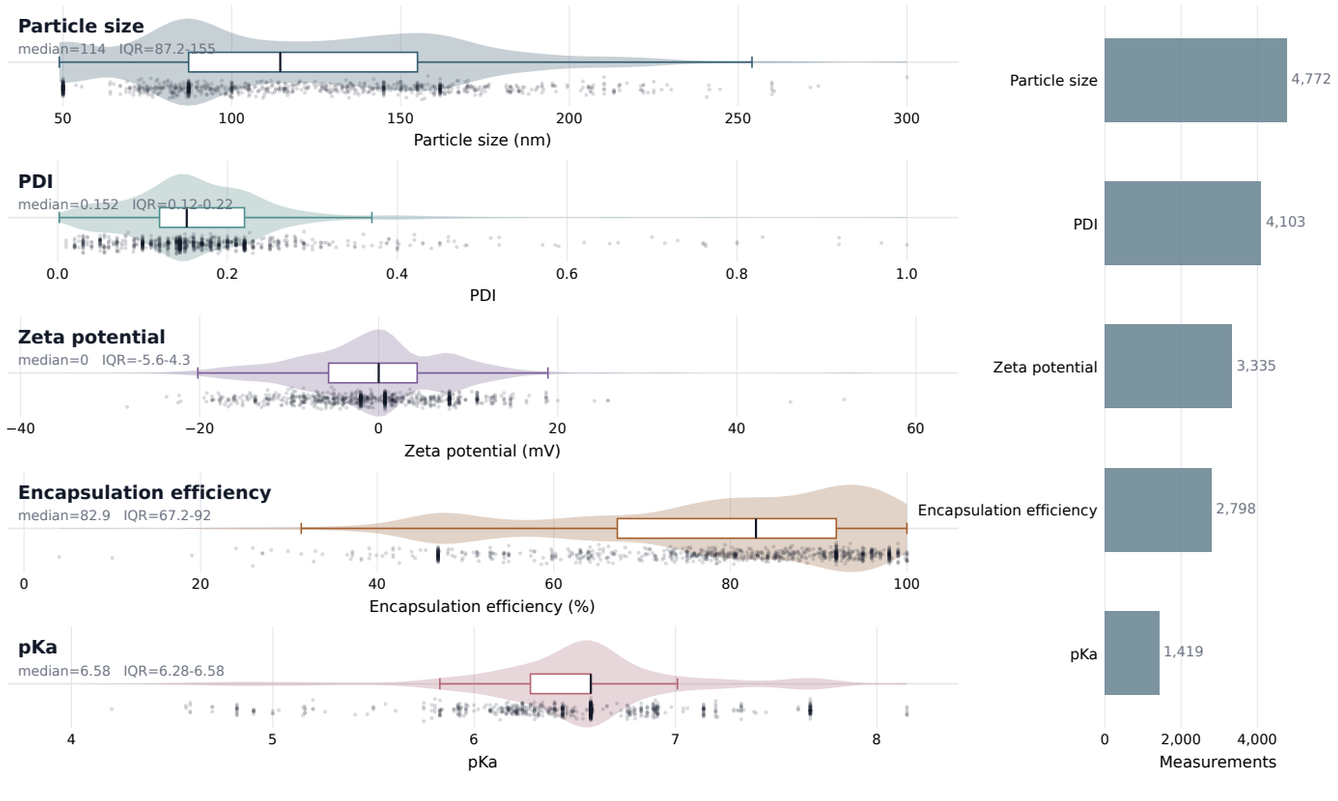

b
Experimental setting
in vitro
20,015
in vivo
3,576
ex vivo
37
0
5,000
10,000
15,000
20,000
Experimental records
Cargo type
mRNA
19,002
siRNA
3,355
pDNA
1,093
0
5,000
10,000
15,000
Experimental records

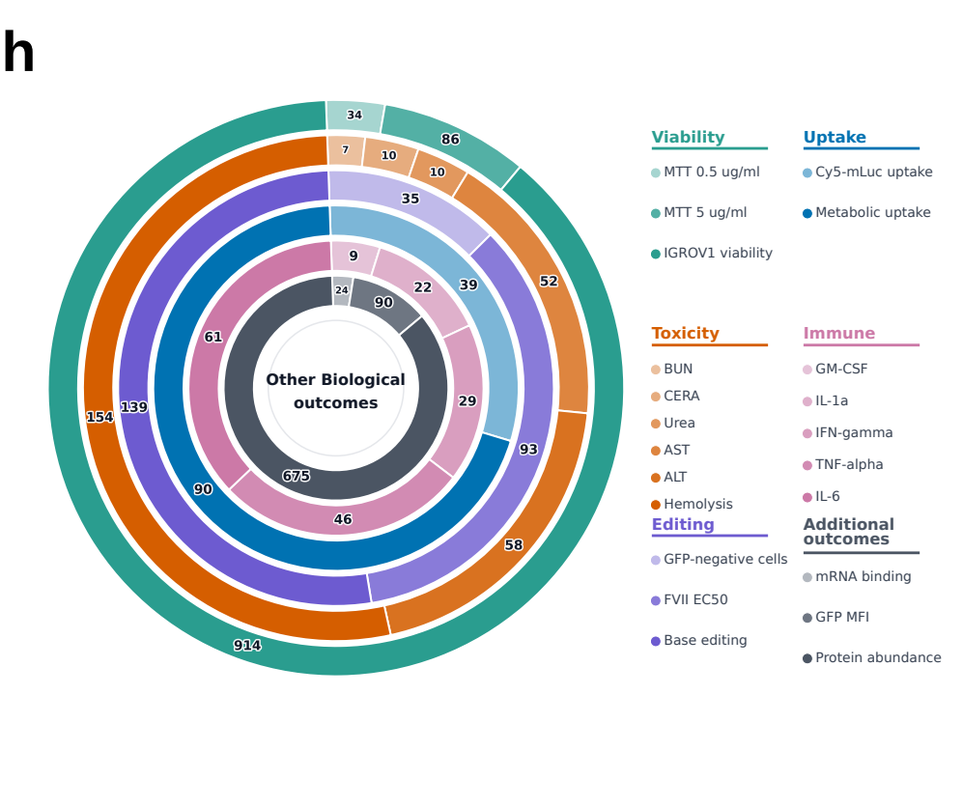

c
studies
cumulative records
cumulative unique ionizable lipids
cumulative unique lnps
Studies
Cumulative count
2010
2016
2019
2022
2025
Publication year derived from study_id


d
Physicochemical properties
Delivery
Viability
Toxicity
Editing
Uptake
Immune response
Other outcomes
Total records
0
2000
Number of records


e
Biological model
HeLa 7,873
A549 2,526
HepG2 2,255
DC2.4 1,843
Mouse C57BL/6 1,814
RAW264.7 1,254
IGROV1 1,206
Other 1,200
BEAS-2B 1,022
Mouse BALBc 751
Mouse C57BL/6J 744
MC38 269
BeWo b30 260
Records
Biological target
lung epithelium 1,836
bronchial epithelium 1,074
muscle 1,003
liver 942
Other 835
multiorgan 331
spleen 286
colon cancer cells 269
lung 152
dendritic cells 140
bone marrow 139
keratinocytes 115
heart 110
Records
Administration route
intravenous 2,295
intramuscular 1,025
topical transdermal 83
intradermal 66
intratracheal 66
inhalation 38
subcutaneous 3
In vivo records


f
Ionizable lipid mol%
n=22,549 median=35
Helper lipid mol%
n=22,451 median=16
Cholesterol mol%
n=22,451 median=46
PEG lipid mol%
n=22,451 median=2.5
0
20
40
60
80
100
Molar ratio (%)


g
Particle size
median=114 IQR=87.2-155
Particle size (nm)
PDI
median=0.152 IQR=0.12-0.22
PDI
Zeta potential
median=0 IQR=-5.6-4.3
Zeta potential (mV)
Encapsulation efficiency
median=82.9 IQR=67.2-92
Encapsulation efficiency (%)
pKa
median=6.58 IQR=6.28-6.58
pKa
Particle size 4,772
PDI 4,103
Zeta potential 3,335
Encapsulation efficiency 2,798
pKa 1,419
Measurements


h
Other Biological outcomes
Viability
MTT 0.5 ug/ml
MTT 5 ug/ml
IGROV1 viability
Uptake
Cy5-mLuc uptake
Metabolic uptake
Toxicity
BUN
CERA
Urea
AST
ALT
Hemolysis
Immune
GM-CSF
IL-1a
IFN-gamma
TNF-alpha
IL-6
Editing
GFP-negative cells
FVII EC50
Base editing
Additional outcomes
mRNA binding
GFP MFI
Protein abundance